



TULSIRAMJI GAIKWAD-PATIL College of Engineering and Technology

Wardha Road, Nagpur - 441108

Accredited with NAAC A+ Grade

Approved by AICTE, New Delhi, Govt. of Maharashtra



(An Autonomous Institute Affiliated to RTM Nagpur University with NAAC A+ Grade)

Department of Information Technology

Session 2025-26(Odd Semester)

VISION	MISSION
To emerge as a learning hub and center of excellence in the domain of Information Technology	[M1] To impart quality technical education through effective teaching learning process. [M2] To provide a platform to address societal issues as well as challenges faced by IT industries. [M3] To foster a culture of research and impart innovative and entrepreneurial skills in the field of IT. [M4] To ensure overall development of students and staff by inculcating knowledge and professional ethics as a part of lifelong learning.

End Semester

Semester: V

Programme: B. Tech Information Technology

Course: Advanced Programming with Java

Time: 3 Hours

Max Marks: 60 Marks

Instructions to candidates:

1. All questions are compulsory in Group A
2. Solve any two sub-question out of three in Group B
3. Assume suitable data wherever necessary
4. All question carries marks as indicated

BIT33501	Course Outcome
CO1	To classify core Java fundamentals.
CO2	To analyze arrays, packages, and interfaces to develop modular and reusable Java applications.
CO3	To develop and manage multithreaded applications.
CO4	To utilize inheritance and polymorphism principles to create scalable Java applications.
CO5	To demonstrate knowledge of JDBC for database connectivity.

Group A				
Que. 1	Answer the following Question	CO	BTL	Marks
a.	Define JVM.	CO1	2	2
b.	Define array.	CO2	2	2
c.	Define thread.	CO	2	2
d.	Define inheritance.	CO	3	2
e.	Define JDBC.	CO	2	2
Group B				
Que. 2	Answer the following Question (Solve any Two)	CO	BTL	Marks

a.	Explain JVM and bytecode.	CO1	2	5
b.	Explain operators and control structures.	CO1	4	5
c.	Explain collections framework.	CO1	3	5
Que 3.	Answer the following Question (Solve any Two)	CO	BTL	Marks
a.	Explain arrays and interfaces.	CO2	2	5
b.	Explain exception handling with program.	CO2	3	5
c.	Explain packages.	CO2	3	5
Que 4.	Answer the following Question (Solve any Two)	CO	BTL	Marks
a.	Explain multithreading concepts.	CO3	2	5
b.	Write program for thread creation.	CO3	4	5
c.	Explain synchronization.	CO3	3	5
Que 5.	Answer the following Question (Solve any Two)	CO	BTL	Marks
a.	Explain inheritance types.	CO4	4	5
b.	Explain overriding vs overloading.	CO4	3	5
c.	Explain super keyword.	CO4	3	5
Que 6.	Answer the following Question (Solve any Two)	CO	BTL	Marks
a.	Explain JDBC architecture.	CO5	2	5
b.	Write steps for DB connection.	CO5	3	5
c.	Explain driver types.	CO5	3	5

Bloom's Taxonomy Level (BTL)- Remember , Understand , Apply, Analyze , Evaluate , Create